IBM Z Datathon: Theme 3 - Wildcard

PROJECT DISPATCH

A solution to New York EMS response times.



THE PROBLEM

13 minutes

AVERAGE URGENT CARE AMBULANCE RESPONSE TIME IN NEW YORK

Legacy infrastructure cannot handle the low staffing and high call volume especially in underserved boroughs.

"PEOPLE ARE DYING BECAUSE THERE ARE NO STAFF,"

Anthony Almojera, a lieutenant paramedic recognising the ever increasing callout time.

"CLIMATE PROJECTIONS FOR NEW YORK CITY SHOWS SOME OF THE STATE'S LARGEST INCREASE IN EXTREME HEAT,"

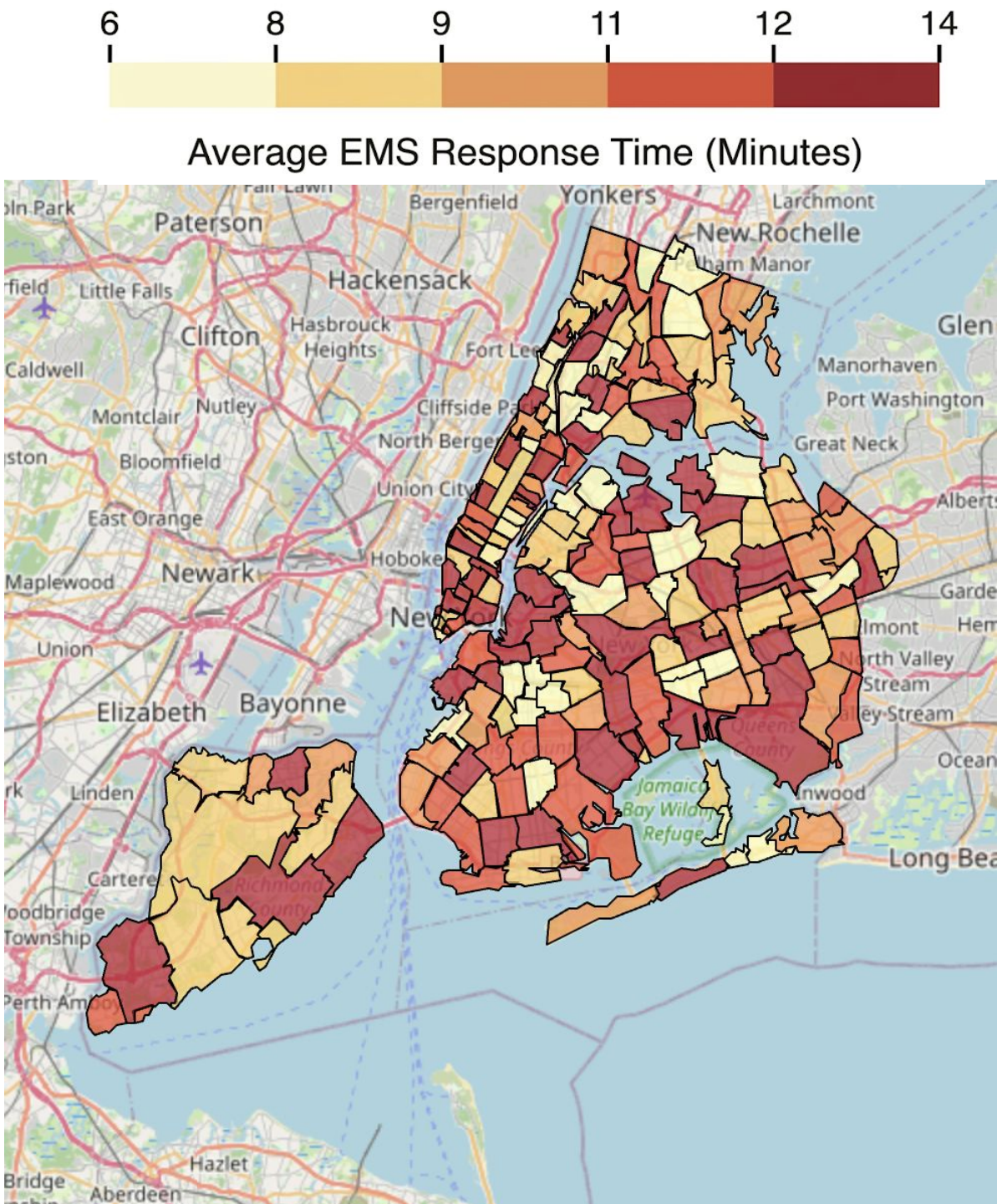
Climate change is creating more drastic weather, which effects the most vulnerable people.

THE TARGET DEMOGRAPHIC

PRIMARY ADOPTION FOCUSED ON DENSE METROPOLITAN REGIONS - OUR TARGET BEING NYC

490,000
VEHICLES

8.8M
NYC POPULATION



THE SOLUTION



MOBILE MEDICAL UNITS

Dynamic route optimization across NYC. The optimised number of units has been calculated to 330.



PREDICTIVE ROUTING

Historical incident-based unit distribution.



INCIDENT PREDICTION

Weather-triggered preventative medical deployment.



PRIORITY LOCATIONS

EMS dispatch data hotspot mapping.

Source: NYC Open Data EMS incident dispatch data; replay of 2022-07-19, 4,694 calls.

THE IMPLEMENTATION

Data-driven methodology built for enterprise emergency operations.

01 / DATA

NYC 911 calls, weather, Census, and hospital data joined by ZIP zone.

1.5M

2021 calls used to train.

02 / FORECAST

LightGBM predicts critical call volume for the next 30 to 180 minutes.

8 Models

Tested on all 2022 data.

03 / POSITION

Idle units preemptively route to expected demand zones. Granite LLM translates output.

30 Min

Repositioning interval.

04 / REPLAY

Calls replayed against a fixed layout using the exact same fleet.

82% → 91%

Critical calls reached < 8 mins.

POWERED BY

IBM LINUXONE

s390x Architecture

Model trained, replayed, and demo served entirely from the VM.

33 / 33

Automated system tests passed successfully on the VM.

Modelled Scale

Assumed 330-unit fleet size (response times are modelled).

THE STACK ON IBM Z

Every layer below runs on the LinuxONE instance, from raw data to the demo.

DEMO
React + Leaflet map, served from the VM; real street routes (OSRM)

DECISIONS
Positioning policy (SciPy assignment) and minute-by-minute replay of real 2022 calls

FORECAST
8 LightGBM Poisson models: critical and other calls per ZIP zone, 30 min to 3 h

DATA AND PYTHON
pandas, NumPy, SciPy; NYC Open Data, Open-Meteo, Census

BASE OF THE STACK

IBM LINUXONE
s390x (big-endian), z17-generation, z/VM 7.3, 3 CPUs, 8 GB, Jupyter

HOW WE USE IBM Z

End to end
Training, scoring, replay and demo all run on the instance. 33 of 33 tests pass on s390x.

1.0 GB
peak memory for two years of data: built to fit the instance, not the other way round.

Portable
Models saved as text files, because pickles do not move between x86 and big-endian s390x.

nnpa
On-chip AI accelerator found and driven through Snap ML. Benchmarked at 0.3 to 0.7x the CPU, so we kept the faster CPU path.

Measured on the VM: accelerator benchmark, memory test and test suite (screenshots in the repo).

PROJECT DISPATCH

A solution to New York EMS response times.

Real calls: Tue, 19 Jul 2022
Response times are modelled

PAUSED
16:08

USER INTERFACE

REPLAY CONTROL

RUN DEMO

PLAY | RESET

Speed: minutes of city time per second

1× | 2× | 4× | 8×

Time 16:08 (129 of 360 min)

VIEW

COMPARE | MODEL | STATIC

Zone shading shows

- MODEL RISK INDEX
- FORECAST CALLS, NEXT HOUR
- WHAT ACTUALLY HAPPENED

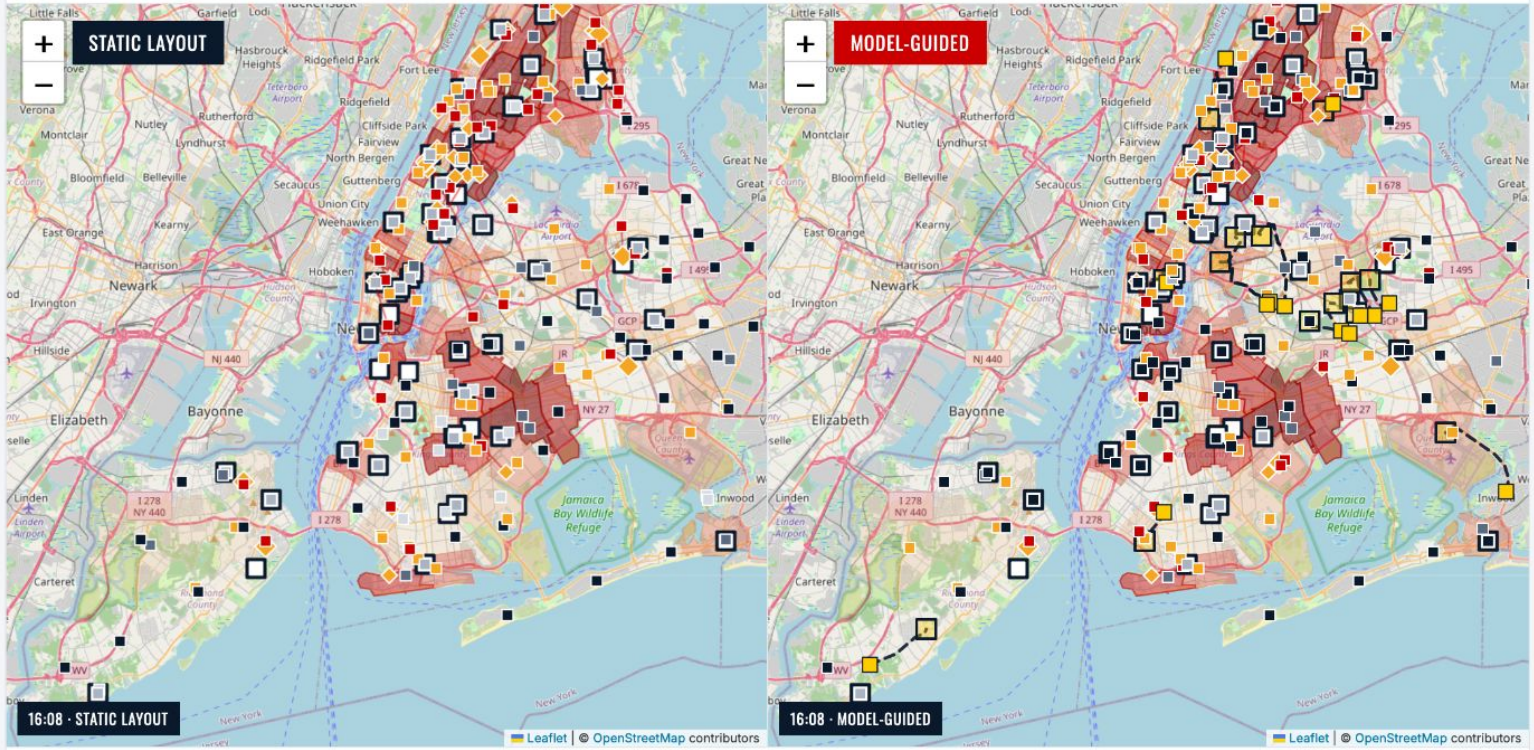
- Zone shading ☑
- Destination lines ☑
- Incoming calls ☑
- Coverage areas ☐

FLEET: 330 UNITS

Available	134
Moving to a new post	15
Responding to a call	28

NYC OPERATIONS MAP

Forecast issued 16:00 · 9 min ago · refreshed every 30 min · shading: model risk index



STATIC LAYOUT

91.8% Critical calls reached within 8 min

4.7 MIN Mean response

MODEL-GUIDED

96.0% Critical calls reached within 8 min

3.9 MIN Mean response

INCOMING CALLS

0	28	502
Waiting	Unit en route	Reached

- 16:08 queens 11419 **critical** — unit en route
- 16:08 bronx 10469 **critical** — unit en route
- 16:08 manhattan 10026 — unit en route
- 16:08 queens 11355 **critical** — unit en route

HIGHEST-RISK ZONES, NEXT HOUR

ZIP ZONE	RISK	STATIC	MODEL
10456 BRO	100	0	2
11212 BRO	99	0	3
10029 MAN	99	0	0
10457 BRO	98	0	0
11207 BRO	98	0	2
10458 BRO	97	0	1
10467 BRO	97	0	1

Units parked in, or driving to, each zone's area. The model moves units toward the zones it expects to be busy.

MODEL DECISIONS

16:00 Unit 0
Sent to post: 11213 to 11225 (brooklyn), 7 min drive. Expected-call coverage +0.41

16:00 Unit 10
Sent to post: 10461 to 10459 (bronx), 19

THE COMPETITIVE ADVANTAGE

LEGACY SOLUTIONS

13 MIN

Mean response time.

- ✕ Static, reactive deployment.

PROJECT DISPATCH

9.65 MIN

Mean response time.

- ✓ 90% of critical calls reached in 8 minutes.
- ✓ Improved coverage in underserved areas.

THE TEAM



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THE FUTURE

BUILDING THE NEXT STANDARD OF URBAN RESPONSE.

Questions & Discussion

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